Device: cuda
GPU: Tesla T4
All 20 labeled DRIVE training images are available.

==========
RUN: fold_1
Training IDs: [21, 24, 25, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 39, 40]
Validation IDs: [22, 23, 26, 38]
Normalization: mean=0.4333, std=0.1060
Training vessel fraction: 0.1271
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fold_1 | Epoch 01/40 | Loss=1.3940 | AUC=0.9442 | Dice=0.7083 | Threshold=0.70 | Accuracy=0.9342 | Sensitivity=0.6748 | Specificity=0.9690 | Precision=0.7453 | 0.1 min
fold_1 | Epoch 02/40 | Loss=1.1654 | AUC=0.9521 | Dice=0.7383 | Threshold=0.80 | Accuracy=0.9405 | Sensitivity=0.7087 | Specificity=0.9716 | Precision=0.7705 | 0.3 min
fold_1 | Epoch 03/40 | Loss=1.0609 | AUC=0.9554 | Dice=0.7407 | Threshold=0.80 | Accuracy=0.9362 | Sensitivity=0.7691 | Specificity=0.9586 | Precision=0.7142 | 0.4 min
fold_1 | Epoch 04/40 | Loss=0.9307 | AUC=0.9631 | Dice=0.7633 | Threshold=0.80 | Accuracy=0.9398 | Sensitivity=0.8188 | Specificity=0.9561 | Precision=0.7148 | 0.5 min
fold_1 | Epoch 05/40 | Loss=0.7807 | AUC=0.9681 | Dice=0.7820 | Threshold=0.80 | Accuracy=0.9459 | Sensitivity=0.8188 | Specificity=0.9630 | Precision=0.7483 | 0.6 min
fold_1 | Epoch 06/40 | Loss=0.6633 | AUC=0.9704 | Dice=0.7851 | Threshold=0.70 | Accuracy=0.9496 | Sensitivity=0.7765 | Specificity=0.9729 | Precision=0.7939 | 0.7 min
fold_1 | Epoch 07/40 | Loss=0.6220 | AUC=0.9672 | Dice=0.7473 | Threshold=0.80 | Accuracy=0.9325 | Sensitivity=0.8428 | Specificity=0.9445 | Precision=0.6712 | 0.8 min
fold_1 | Epoch 08/40 | Loss=0.6023 | AUC=0.9712 | Dice=0.7771 | Threshold=0.80 | Accuracy=0.9458 | Sensitivity=0.7978 | Specificity=0.9657 | Precision=0.7574 | 0.9 min
fold_1 | Epoch 09/40 | Loss=0.5810 | AUC=0.9711 | Dice=0.7684 | Threshold=0.80 | Accuracy=0.9401 | Sensitivity=0.8385 | Specificity=0.9538 | Precision=0.7092 | 1.0 min
fold_1 | Epoch 10/40 | Loss=0.5649 | AUC=0.9765 | Dice=0.8026 | Threshold=0.75 | Accuracy=0.9530 | Sensitivity=0.8072 | Specificity=0.9725 | Precision=0.7980 | 1.1 min
fold_1 | Epoch 11/40 | Loss=0.5551 | AUC=0.9728 | Dice=0.7773 | Threshold=0.80 | Accuracy=0.9445 | Sensitivity=0.8173 | Specificity=0.9616 | Precision=0.7410 | 1.2 min
fold_1 | Epoch 12/40 | Loss=0.5554 | AUC=0.9744 | Dice=0.7943 | Threshold=0.80 | Accuracy=0.9508 | Sensitivity=0.8021 | Specificity=0.9708 | Precision=0.7866 | 1.3 min
fold_1 | Epoch 13/40 | Loss=0.5509 | AUC=0.9741 | Dice=0.7841 | Threshold=0.80 | Accuracy=0.9470 | Sensitivity=0.8117 | Specificity=0.9652 | Precision=0.7583 | 1.4 min
fold_1 | Epoch 14/40 | Loss=0.5383 | AUC=0.9764 | Dice=0.7982 | Threshold=0.80 | Accuracy=0.9524 | Sensitivity=0.7941 | Specificity=0.9737 | Precision=0.8024 | 1.5 min
fold_1 | Epoch 15/40 | Loss=0.5378 | AUC=0.9771 | Dice=0.7981 | Threshold=0.80 | Accuracy=0.9501 | Sensitivity=0.8328 | Specificity=0.9658 | Precision=0.7662 | 1.6 min

fold_1 | Epoch 16/40 | Loss=0.5330 | AUC=0.9779 | Dice=0.8045 | Threshold=0.75 |
Accuracy=0.9533 | Sensitivity=0.8114 | Specificity=0.9724 | Precision=0.7978 | 1.7 min
fold_1 | Epoch 17/40 | Loss=0.5356 | AUC=0.9786 | Dice=0.8094 | Threshold=0.75 |
Accuracy=0.9549 | Sensitivity=0.8080 | Specificity=0.9746 | Precision=0.8107 | 1.8 min
fold_1 | Epoch 18/40 | Loss=0.5297 | AUC=0.9782 | Dice=0.8059 | Threshold=0.80 |
Accuracy=0.9541 | Sensitivity=0.8051 | Specificity=0.9741 | Precision=0.8068 | 1.9 min
fold_1 | Epoch 19/40 | Loss=0.5281 | AUC=0.9792 | Dice=0.8092 | Threshold=0.75 |
Accuracy=0.9543 | Sensitivity=0.8178 | Specificity=0.9726 | Precision=0.8007 | 2.0 min
fold_1 | Epoch 20/40 | Loss=0.5246 | AUC=0.9794 | Dice=0.8111 | Threshold=0.75 |
Accuracy=0.9554 | Sensitivity=0.8073 | Specificity=0.9753 | Precision=0.8149 | 2.1 min
fold_1 | Epoch 21/40 | Loss=0.5193 | AUC=0.9762 | Dice=0.7971 | Threshold=0.80 |
Accuracy=0.9517 | Sensitivity=0.8005 | Specificity=0.9720 | Precision=0.7938 | 2.2 min
fold_1 | Epoch 22/40 | Loss=0.5103 | AUC=0.9780 | Dice=0.8029 | Threshold=0.80 |
Accuracy=0.9535 | Sensitivity=0.7988 | Specificity=0.9743 | Precision=0.8071 | 2.3 min
fold_1 | Epoch 23/40 | Loss=0.5163 | AUC=0.9769 | Dice=0.7998 | Threshold=0.70 |
Accuracy=0.9523 | Sensitivity=0.8038 | Specificity=0.9723 | Precision=0.7958 | 2.4 min
fold_1 | Epoch 24/40 | Loss=0.5162 | AUC=0.9774 | Dice=0.7996 | Threshold=0.80 |
Accuracy=0.9520 | Sensitivity=0.8078 | Specificity=0.9714 | Precision=0.7916 | 2.5 min
fold_1 | Epoch 25/40 | Loss=0.5125 | AUC=0.9784 | Dice=0.8060 | Threshold=0.75 |
Accuracy=0.9538 | Sensitivity=0.8105 | Specificity=0.9730 | Precision=0.8015 | 2.6 min
fold_1 | Epoch 26/40 | Loss=0.5116 | AUC=0.9779 | Dice=0.8003 | Threshold=0.80 |
Accuracy=0.9520 | Sensitivity=0.8117 | Specificity=0.9708 | Precision=0.7891 | 2.7 min
fold_1 | Epoch 27/40 | Loss=0.5094 | AUC=0.9784 | Dice=0.8051 | Threshold=0.80 |
Accuracy=0.9540 | Sensitivity=0.8020 | Specificity=0.9744 | Precision=0.8083 | 2.8 min
fold_1 | Epoch 28/40 | Loss=0.5058 | AUC=0.9792 | Dice=0.8103 | Threshold=0.75 |
Accuracy=0.9549 | Sensitivity=0.8138 | Specificity=0.9738 | Precision=0.8069 | 2.9 min
Early stopping: validation Dice did not improve for 8 epochs.
Saved checkpoint: /content/vessel_model_5fold/fold_1_best.pt
Saved report: /content/vessel_model_5fold/fold_1_report.json

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RUN: fold_2
Training IDs: [22, 23, 24, 25, 26, 28, 29, 30, 31, 33, 34, 35, 36, 37, 38, 40]
Validation IDs: [21, 27, 32, 39]
Normalization: mean=0.4270, std=0.1081
Training vessel fraction: 0.1267
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fold_2 | Epoch 01/40 | Loss=1.3738 | AUC=0.9483 | Dice=0.7080 | Threshold=0.65 |
Accuracy=0.9290 | Sensitivity=0.7152 | Specificity=0.9582 | Precision=0.7009 | 0.1 min
fold_2 | Epoch 02/40 | Loss=1.1464 | AUC=0.9580 | Dice=0.7600 | Threshold=0.65 |
Accuracy=0.9422 | Sensitivity=0.7602 | Specificity=0.9671 | Precision=0.7598 | 0.2 min

fold_2 | Epoch 03/40 | Loss=1.0167 | AUC=0.9671 | Dice=0.7801 | Threshold=0.75 | Accuracy=0.9464 | Sensitivity=0.7894 | Specificity=0.9679 | Precision=0.7710 | 0.3 min
fold_2 | Epoch 04/40 | Loss=0.8674 | AUC=0.9719 | Dice=0.7920 | Threshold=0.80 | Accuracy=0.9475 | Sensitivity=0.8306 | Specificity=0.9635 | Precision=0.7569 | 0.4 min
fold_2 | Epoch 05/40 | Loss=0.7360 | AUC=0.9730 | Dice=0.7912 | Threshold=0.80 | Accuracy=0.9461 | Sensitivity=0.8485 | Specificity=0.9594 | Precision=0.7411 | 0.5 min
fold_2 | Epoch 06/40 | Loss=0.6580 | AUC=0.9762 | Dice=0.8012 | Threshold=0.70 | Accuracy=0.9514 | Sensitivity=0.8143 | Specificity=0.9701 | Precision=0.7886 | 0.6 min
fold_2 | Epoch 07/40 | Loss=0.6217 | AUC=0.9790 | Dice=0.8095 | Threshold=0.80 | Accuracy=0.9521 | Sensitivity=0.8457 | Specificity=0.9666 | Precision=0.7762 | 0.7 min
fold_2 | Epoch 08/40 | Loss=0.6107 | AUC=0.9794 | Dice=0.8111 | Threshold=0.80 | Accuracy=0.9523 | Sensitivity=0.8509 | Specificity=0.9662 | Precision=0.7749 | 0.8 min
fold_2 | Epoch 09/40 | Loss=0.5948 | AUC=0.9798 | Dice=0.8129 | Threshold=0.80 | Accuracy=0.9529 | Sensitivity=0.8504 | Specificity=0.9669 | Precision=0.7785 | 0.9 min
fold_2 | Epoch 10/40 | Loss=0.5836 | AUC=0.9795 | Dice=0.8119 | Threshold=0.75 | Accuracy=0.9536 | Sensitivity=0.8322 | Specificity=0.9702 | Precision=0.7927 | 1.0 min
fold_2 | Epoch 11/40 | Loss=0.5753 | AUC=0.9814 | Dice=0.8048 | Threshold=0.80 | Accuracy=0.9479 | Sensitivity=0.8934 | Specificity=0.9553 | Precision=0.7322 | 1.1 min
fold_2 | Epoch 12/40 | Loss=0.5661 | AUC=0.9808 | Dice=0.8173 | Threshold=0.80 | Accuracy=0.9545 | Sensitivity=0.8452 | Specificity=0.9695 | Precision=0.7912 | 1.2 min
fold_2 | Epoch 13/40 | Loss=0.5614 | AUC=0.9814 | Dice=0.8207 | Threshold=0.75 | Accuracy=0.9561 | Sensitivity=0.8348 | Specificity=0.9727 | Precision=0.8070 | 1.3 min
fold_2 | Epoch 14/40 | Loss=0.5620 | AUC=0.9814 | Dice=0.8099 | Threshold=0.80 | Accuracy=0.9504 | Sensitivity=0.8771 | Specificity=0.9605 | Precision=0.7522 | 1.4 min
fold_2 | Epoch 15/40 | Loss=0.5480 | AUC=0.9817 | Dice=0.8128 | Threshold=0.80 | Accuracy=0.9510 | Sensitivity=0.8845 | Specificity=0.9601 | Precision=0.7519 | 1.5 min
fold_2 | Epoch 16/40 | Loss=0.5432 | AUC=0.9814 | Dice=0.8139 | Threshold=0.80 | Accuracy=0.9519 | Sensitivity=0.8744 | Specificity=0.9625 | Precision=0.7613 | 1.6 min
fold_2 | Epoch 17/40 | Loss=0.5471 | AUC=0.9823 | Dice=0.8214 | Threshold=0.80 | Accuracy=0.9556 | Sensitivity=0.8478 | Specificity=0.9704 | Precision=0.7965 | 1.7 min
fold_2 | Epoch 18/40 | Loss=0.5497 | AUC=0.9822 | Dice=0.8186 | Threshold=0.80 | Accuracy=0.9530 | Sensitivity=0.8816 | Specificity=0.9627 | Precision=0.7640 | 1.8 min
fold_2 | Epoch 19/40 | Loss=0.5310 | AUC=0.9826 | Dice=0.8164 | Threshold=0.80 | Accuracy=0.9526 | Sensitivity=0.8750 | Specificity=0.9633 | Precision=0.7652 | 1.9 min
fold_2 | Epoch 20/40 | Loss=0.5371 | AUC=0.9826 | Dice=0.8217 | Threshold=0.80 | Accuracy=0.9554 | Sensitivity=0.8545 | Specificity=0.9692 | Precision=0.7914 | 2.0 min
fold_2 | Epoch 21/40 | Loss=0.5359 | AUC=0.9826 | Dice=0.8225 | Threshold=0.80 | Accuracy=0.9567 | Sensitivity=0.8331 | Specificity=0.9736 | Precision=0.8121 | 2.1 min
fold_2 | Epoch 22/40 | Loss=0.5293 | AUC=0.9833 | Dice=0.8260 | Threshold=0.80 | Accuracy=0.9570 | Sensitivity=0.8476 | Specificity=0.9720 | Precision=0.8054 | 2.2 min
fold_2 | Epoch 23/40 | Loss=0.5303 | AUC=0.9822 | Dice=0.8216 | Threshold=0.80 | Accuracy=0.9559 | Sensitivity=0.8444 | Specificity=0.9711 | Precision=0.8000 | 2.3 min
fold_2 | Epoch 24/40 | Loss=0.5224 | AUC=0.9834 | Dice=0.8228 | Threshold=0.80 | Accuracy=0.9549 | Sensitivity=0.8706 | Specificity=0.9664 | Precision=0.7801 | 2.4 min

fold_2 | Epoch 25/40 | Loss=0.5287 | AUC=0.9829 | Dice=0.8218 | Threshold=0.80 |
Accuracy=0.9548 | Sensitivity=0.8657 | Specificity=0.9670 | Precision=0.7821 | 2.5 min
fold_2 | Epoch 26/40 | Loss=0.5206 | AUC=0.9833 | Dice=0.8228 | Threshold=0.80 |
Accuracy=0.9551 | Sensitivity=0.8668 | Specificity=0.9671 | Precision=0.7830 | 2.6 min
fold_2 | Epoch 27/40 | Loss=0.5124 | AUC=0.9836 | Dice=0.8266 | Threshold=0.80 |
Accuracy=0.9577 | Sensitivity=0.8378 | Specificity=0.9741 | Precision=0.8156 | 2.7 min
fold_2 | Epoch 28/40 | Loss=0.5147 | AUC=0.9835 | Dice=0.8177 | Threshold=0.80 |
Accuracy=0.9527 | Sensitivity=0.8806 | Specificity=0.9626 | Precision=0.7632 | 2.8 min
fold_2 | Epoch 29/40 | Loss=0.5174 | AUC=0.9835 | Dice=0.8206 | Threshold=0.80 |
Accuracy=0.9541 | Sensitivity=0.8724 | Specificity=0.9653 | Precision=0.7746 | 2.9 min
fold_2 | Epoch 30/40 | Loss=0.5092 | AUC=0.9839 | Dice=0.8254 | Threshold=0.80 |
Accuracy=0.9559 | Sensitivity=0.8663 | Specificity=0.9682 | Precision=0.7882 | 3.0 min
fold_2 | Epoch 31/40 | Loss=0.5048 | AUC=0.9841 | Dice=0.8255 | Threshold=0.80 |
Accuracy=0.9559 | Sensitivity=0.8665 | Specificity=0.9681 | Precision=0.7881 | 3.1 min
fold_2 | Epoch 32/40 | Loss=0.5053 | AUC=0.9841 | Dice=0.8251 | Threshold=0.80 |
Accuracy=0.9557 | Sensitivity=0.8677 | Specificity=0.9678 | Precision=0.7865 | 3.2 min
fold_2 | Epoch 33/40 | Loss=0.5068 | AUC=0.9841 | Dice=0.8252 | Threshold=0.80 |
Accuracy=0.9565 | Sensitivity=0.8526 | Specificity=0.9707 | Precision=0.7995 | 3.3 min
fold_2 | Epoch 34/40 | Loss=0.5061 | AUC=0.9840 | Dice=0.8276 | Threshold=0.80 |
Accuracy=0.9576 | Sensitivity=0.8451 | Specificity=0.9730 | Precision=0.8109 | 3.4 min
fold_2 | Epoch 35/40 | Loss=0.4954 | AUC=0.9842 | Dice=0.8259 | Threshold=0.80 |
Accuracy=0.9560 | Sensitivity=0.8661 | Specificity=0.9683 | Precision=0.7892 | 3.5 min
fold_2 | Epoch 36/40 | Loss=0.5088 | AUC=0.9835 | Dice=0.8260 | Threshold=0.80 |
Accuracy=0.9563 | Sensitivity=0.8627 | Specificity=0.9691 | Precision=0.7923 | 3.6 min
fold_2 | Epoch 37/40 | Loss=0.5015 | AUC=0.9841 | Dice=0.8267 | Threshold=0.80 |
Accuracy=0.9565 | Sensitivity=0.8611 | Specificity=0.9696 | Precision=0.7949 | 3.7 min
fold_2 | Epoch 38/40 | Loss=0.4994 | AUC=0.9843 | Dice=0.8267 | Threshold=0.80 |
Accuracy=0.9565 | Sensitivity=0.8611 | Specificity=0.9696 | Precision=0.7949 | 3.8 min
fold_2 | Epoch 39/40 | Loss=0.5004 | AUC=0.9842 | Dice=0.8267 | Threshold=0.80 |
Accuracy=0.9567 | Sensitivity=0.8585 | Specificity=0.9701 | Precision=0.7970 | 3.9 min
fold_2 | Epoch 40/40 | Loss=0.4971 | AUC=0.9840 | Dice=0.8272 | Threshold=0.80 |
Accuracy=0.9571 | Sensitivity=0.8535 | Specificity=0.9712 | Precision=0.8024 | 4.0 min
Saved checkpoint: /content/vessel_model_5fold/fold_2_best.pt
Saved report: /content/vessel_model_5fold/fold_2_report.json

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RUN: fold_3
Training IDs: [21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 32, 33, 36, 37, 38, 39]
Validation IDs: [31, 34, 35, 40]
Normalization: mean=0.4419, std=0.1022
Training vessel fraction: 0.1276
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fold_3 | Epoch 01/40 | Loss=1.4484 | AUC=0.9252 | Dice=0.6782 | Threshold=0.80 | Accuracy=0.9253 | Sensitivity=0.6765 | Specificity=0.9580 | Precision=0.6799 | 0.1 min
fold_3 | Epoch 02/40 | Loss=1.2439 | AUC=0.9342 | Dice=0.7061 | Threshold=0.80 | Accuracy=0.9337 | Sensitivity=0.6846 | Specificity=0.9665 | Precision=0.7289 | 0.2 min
fold_3 | Epoch 03/40 | Loss=1.1139 | AUC=0.9530 | Dice=0.7424 | Threshold=0.80 | Accuracy=0.9417 | Sensitivity=0.7214 | Specificity=0.9707 | Precision=0.7646 | 0.3 min
fold_3 | Epoch 04/40 | Loss=0.9384 | AUC=0.9608 | Dice=0.7582 | Threshold=0.80 | Accuracy=0.9426 | Sensitivity=0.7729 | Specificity=0.9650 | Precision=0.7441 | 0.4 min
fold_3 | Epoch 05/40 | Loss=0.7605 | AUC=0.9513 | Dice=0.7345 | Threshold=0.80 | Accuracy=0.9380 | Sensitivity=0.7369 | Specificity=0.9645 | Precision=0.7321 | 0.5 min
fold_3 | Epoch 06/40 | Loss=0.6454 | AUC=0.9642 | Dice=0.7680 | Threshold=0.80 | Accuracy=0.9460 | Sensitivity=0.7684 | Specificity=0.9694 | Precision=0.7677 | 0.6 min
fold_3 | Epoch 07/40 | Loss=0.5893 | AUC=0.9580 | Dice=0.7556 | Threshold=0.75 | Accuracy=0.9432 | Sensitivity=0.7547 | Specificity=0.9680 | Precision=0.7566 | 0.7 min
fold_3 | Epoch 08/40 | Loss=0.5694 | AUC=0.9585 | Dice=0.7475 | Threshold=0.80 | Accuracy=0.9419 | Sensitivity=0.7383 | Specificity=0.9688 | Precision=0.7569 | 0.8 min
fold_3 | Epoch 09/40 | Loss=0.5509 | AUC=0.9643 | Dice=0.7664 | Threshold=0.80 | Accuracy=0.9456 | Sensitivity=0.7661 | Specificity=0.9693 | Precision=0.7667 | 0.9 min
fold_3 | Epoch 10/40 | Loss=0.5475 | AUC=0.9629 | Dice=0.7669 | Threshold=0.80 | Accuracy=0.9444 | Sensitivity=0.7864 | Specificity=0.9652 | Precision=0.7484 | 1.0 min
fold_3 | Epoch 11/40 | Loss=0.5395 | AUC=0.9624 | Dice=0.7619 | Threshold=0.75 | Accuracy=0.9448 | Sensitivity=0.7585 | Specificity=0.9694 | Precision=0.7654 | 1.1 min
fold_3 | Epoch 12/40 | Loss=0.5355 | AUC=0.9601 | Dice=0.7615 | Threshold=0.80 | Accuracy=0.9458 | Sensitivity=0.7438 | Specificity=0.9724 | Precision=0.7801 | 1.2 min
fold_3 | Epoch 13/40 | Loss=0.5226 | AUC=0.9599 | Dice=0.7689 | Threshold=0.75 | Accuracy=0.9473 | Sensitivity=0.7537 | Specificity=0.9727 | Precision=0.7847 | 1.3 min
fold_3 | Epoch 14/40 | Loss=0.5249 | AUC=0.9692 | Dice=0.7809 | Threshold=0.80 | Accuracy=0.9482 | Sensitivity=0.7935 | Specificity=0.9686 | Precision=0.7688 | 1.4 min
fold_3 | Epoch 15/40 | Loss=0.5184 | AUC=0.9570 | Dice=0.7577 | Threshold=0.80 | Accuracy=0.9455 | Sensitivity=0.7320 | Specificity=0.9736 | Precision=0.7853 | 1.5 min
fold_3 | Epoch 16/40 | Loss=0.5147 | AUC=0.9716 | Dice=0.7860 | Threshold=0.80 | Accuracy=0.9490 | Sensitivity=0.8039 | Specificity=0.9682 | Precision=0.7688 | 1.6 min
fold_3 | Epoch 17/40 | Loss=0.5142 | AUC=0.9636 | Dice=0.7701 | Threshold=0.80 | Accuracy=0.9461 | Sensitivity=0.7752 | Specificity=0.9687 | Precision=0.7652 | 1.7 min
fold_3 | Epoch 18/40 | Loss=0.5056 | AUC=0.9528 | Dice=0.7621 | Threshold=0.80 | Accuracy=0.9451 | Sensitivity=0.7553 | Specificity=0.9701 | Precision=0.7689 | 1.8 min
fold_3 | Epoch 19/40 | Loss=0.5059 | AUC=0.9689 | Dice=0.7793 | Threshold=0.80 | Accuracy=0.9493 | Sensitivity=0.7697 | Specificity=0.9729 | Precision=0.7891 | 1.9 min
fold_3 | Epoch 20/40 | Loss=0.5119 | AUC=0.9733 | Dice=0.7923 | Threshold=0.80 | Accuracy=0.9508 | Sensitivity=0.8070 | Specificity=0.9697 | Precision=0.7781 | 2.0 min
fold_3 | Epoch 21/40 | Loss=0.5018 | AUC=0.9519 | Dice=0.7687 | Threshold=0.75 | Accuracy=0.9469 | Sensitivity=0.7577 | Specificity=0.9718 | Precision=0.7800 | 2.1 min
fold_3 | Epoch 22/40 | Loss=0.5067 | AUC=0.9625 | Dice=0.7699 | Threshold=0.80 | Accuracy=0.9467 | Sensitivity=0.7659 | Specificity=0.9705 | Precision=0.7740 | 2.2 min

fold_3 | Epoch 23/40 | Loss=0.4930 | AUC=0.9555 | Dice=0.7680 | Threshold=0.80 |
Accuracy=0.9464 | Sensitivity=0.7619 | Specificity=0.9707 | Precision=0.7742 | 2.3 min
fold_3 | Epoch 24/40 | Loss=0.4998 | AUC=0.9661 | Dice=0.7765 | Threshold=0.80 |
Accuracy=0.9481 | Sensitivity=0.7754 | Specificity=0.9708 | Precision=0.7777 | 2.4 min
fold_3 | Epoch 25/40 | Loss=0.4913 | AUC=0.9478 | Dice=0.7590 | Threshold=0.80 |
Accuracy=0.9461 | Sensitivity=0.7286 | Specificity=0.9748 | Precision=0.7921 | 2.5 min
fold_3 | Epoch 26/40 | Loss=0.4875 | AUC=0.9435 | Dice=0.7593 | Threshold=0.80 |
Accuracy=0.9452 | Sensitivity=0.7424 | Specificity=0.9719 | Precision=0.7771 | 2.6 min
fold_3 | Epoch 27/40 | Loss=0.4928 | AUC=0.9404 | Dice=0.7610 | Threshold=0.80 |
Accuracy=0.9462 | Sensitivity=0.7355 | Specificity=0.9740 | Precision=0.7884 | 2.7 min
fold_3 | Epoch 28/40 | Loss=0.4872 | AUC=0.9609 | Dice=0.7703 | Threshold=0.80 |
Accuracy=0.9459 | Sensitivity=0.7799 | Specificity=0.9677 | Precision=0.7610 | 2.8 min
Early stopping: validation Dice did not improve for 8 epochs.
Saved checkpoint: /content/vessel_model_5fold/fold_3_best.pt
Saved report: /content/vessel_model_5fold/fold_3_report.json

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RUN: fold_4
Training IDs: [21, 22, 23, 24, 25, 26, 27, 28, 31, 32, 34, 35, 36, 38, 39, 40]
Validation IDs: [29, 30, 33, 37]
Normalization: mean=0.4457, std=0.1069
Training vessel fraction: 0.1267
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fold_4 | Epoch 01/40 | Loss=1.4097 | AUC=0.9268 | Dice=0.6829 | Threshold=0.65 |
Accuracy=0.9220 | Sensitivity=0.6998 | Specificity=0.9523 | Precision=0.6668 | 0.1 min
fold_4 | Epoch 02/40 | Loss=1.1832 | AUC=0.9549 | Dice=0.7599 | Threshold=0.75 |
Accuracy=0.9424 | Sensitivity=0.7598 | Specificity=0.9673 | Precision=0.7600 | 0.2 min
fold_4 | Epoch 03/40 | Loss=1.0466 | AUC=0.9598 | Dice=0.7750 | Threshold=0.75 |
Accuracy=0.9458 | Sensitivity=0.7771 | Specificity=0.9689 | Precision=0.7728 | 0.3 min
fold_4 | Epoch 04/40 | Loss=0.8965 | AUC=0.9642 | Dice=0.7828 | Threshold=0.70 |
Accuracy=0.9474 | Sensitivity=0.7894 | Specificity=0.9690 | Precision=0.7762 | 0.4 min
fold_4 | Epoch 05/40 | Loss=0.7444 | AUC=0.9696 | Dice=0.7977 | Threshold=0.75 |
Accuracy=0.9519 | Sensitivity=0.7907 | Specificity=0.9738 | Precision=0.8048 | 0.5 min
fold_4 | Epoch 06/40 | Loss=0.6603 | AUC=0.9712 | Dice=0.8026 | Threshold=0.80 |
Accuracy=0.9531 | Sensitivity=0.7950 | Specificity=0.9746 | Precision=0.8104 | 0.6 min
fold_4 | Epoch 07/40 | Loss=0.6216 | AUC=0.9746 | Dice=0.8026 | Threshold=0.75 |
Accuracy=0.9522 | Sensitivity=0.8102 | Specificity=0.9715 | Precision=0.7952 | 0.7 min
fold_4 | Epoch 08/40 | Loss=0.5879 | AUC=0.9764 | Dice=0.8118 | Threshold=0.65 |
Accuracy=0.9550 | Sensitivity=0.8084 | Specificity=0.9750 | Precision=0.8151 | 0.8 min
fold_4 | Epoch 09/40 | Loss=0.5774 | AUC=0.9739 | Dice=0.8025 | Threshold=0.80 |
Accuracy=0.9503 | Sensitivity=0.8416 | Specificity=0.9651 | Precision=0.7669 | 0.9 min

fold_4 | Epoch 10/40 | Loss=0.5715 | AUC=0.9771 | Dice=0.8092 | Threshold=0.75 | Accuracy=0.9545 | Sensitivity=0.8042 | Specificity=0.9750 | Precision=0.8142 | 1.0 min
fold_4 | Epoch 11/40 | Loss=0.5677 | AUC=0.9776 | Dice=0.8162 | Threshold=0.80 | Accuracy=0.9558 | Sensitivity=0.8172 | Specificity=0.9747 | Precision=0.8151 | 1.1 min
fold_4 | Epoch 12/40 | Loss=0.5619 | AUC=0.9776 | Dice=0.8092 | Threshold=0.80 | Accuracy=0.9540 | Sensitivity=0.8139 | Specificity=0.9731 | Precision=0.8046 | 1.2 min
fold_4 | Epoch 13/40 | Loss=0.5548 | AUC=0.9785 | Dice=0.8140 | Threshold=0.75 | Accuracy=0.9551 | Sensitivity=0.8185 | Specificity=0.9737 | Precision=0.8095 | 1.3 min
fold_4 | Epoch 14/40 | Loss=0.5534 | AUC=0.9783 | Dice=0.8141 | Threshold=0.70 | Accuracy=0.9551 | Sensitivity=0.8191 | Specificity=0.9736 | Precision=0.8091 | 1.4 min
fold_4 | Epoch 15/40 | Loss=0.5498 | AUC=0.9782 | Dice=0.8163 | Threshold=0.75 | Accuracy=0.9560 | Sensitivity=0.8137 | Specificity=0.9754 | Precision=0.8188 | 1.5 min
fold_4 | Epoch 16/40 | Loss=0.5536 | AUC=0.9787 | Dice=0.8184 | Threshold=0.75 | Accuracy=0.9558 | Sensitivity=0.8297 | Specificity=0.9730 | Precision=0.8075 | 1.6 min
fold_4 | Epoch 17/40 | Loss=0.5349 | AUC=0.9785 | Dice=0.8159 | Threshold=0.80 | Accuracy=0.9560 | Sensitivity=0.8129 | Specificity=0.9755 | Precision=0.8190 | 1.7 min
fold_4 | Epoch 18/40 | Loss=0.5378 | AUC=0.9787 | Dice=0.8155 | Threshold=0.70 | Accuracy=0.9558 | Sensitivity=0.8143 | Specificity=0.9751 | Precision=0.8168 | 1.8 min
fold_4 | Epoch 19/40 | Loss=0.5263 | AUC=0.9798 | Dice=0.8226 | Threshold=0.75 | Accuracy=0.9570 | Sensitivity=0.8308 | Specificity=0.9742 | Precision=0.8145 | 1.9 min
fold_4 | Epoch 20/40 | Loss=0.5367 | AUC=0.9795 | Dice=0.8192 | Threshold=0.75 | Accuracy=0.9564 | Sensitivity=0.8225 | Specificity=0.9747 | Precision=0.8158 | 2.0 min
fold_4 | Epoch 21/40 | Loss=0.5320 | AUC=0.9795 | Dice=0.8195 | Threshold=0.80 | Accuracy=0.9557 | Sensitivity=0.8377 | Specificity=0.9718 | Precision=0.8020 | 2.1 min
fold_4 | Epoch 22/40 | Loss=0.5235 | AUC=0.9805 | Dice=0.8209 | Threshold=0.75 | Accuracy=0.9568 | Sensitivity=0.8257 | Specificity=0.9746 | Precision=0.8162 | 2.2 min
fold_4 | Epoch 23/40 | Loss=0.5157 | AUC=0.9806 | Dice=0.8236 | Threshold=0.75 | Accuracy=0.9571 | Sensitivity=0.8346 | Specificity=0.9738 | Precision=0.8128 | 2.3 min
fold_4 | Epoch 24/40 | Loss=0.5147 | AUC=0.9804 | Dice=0.8215 | Threshold=0.75 | Accuracy=0.9570 | Sensitivity=0.8248 | Specificity=0.9750 | Precision=0.8182 | 2.4 min
fold_4 | Epoch 25/40 | Loss=0.5165 | AUC=0.9804 | Dice=0.8207 | Threshold=0.75 | Accuracy=0.9568 | Sensitivity=0.8243 | Specificity=0.9748 | Precision=0.8171 | 2.5 min
fold_4 | Epoch 26/40 | Loss=0.5215 | AUC=0.9807 | Dice=0.8226 | Threshold=0.75 | Accuracy=0.9571 | Sensitivity=0.8295 | Specificity=0.9745 | Precision=0.8158 | 2.6 min
fold_4 | Epoch 27/40 | Loss=0.5156 | AUC=0.9806 | Dice=0.8219 | Threshold=0.80 | Accuracy=0.9573 | Sensitivity=0.8210 | Specificity=0.9759 | Precision=0.8227 | 2.7 min
fold_4 | Epoch 28/40 | Loss=0.5117 | AUC=0.9813 | Dice=0.8238 | Threshold=0.75 | Accuracy=0.9573 | Sensitivity=0.8316 | Specificity=0.9745 | Precision=0.8162 | 2.8 min
fold_4 | Epoch 29/40 | Loss=0.5103 | AUC=0.9806 | Dice=0.8241 | Threshold=0.75 | Accuracy=0.9574 | Sensitivity=0.8317 | Specificity=0.9745 | Precision=0.8165 | 2.9 min
fold_4 | Epoch 30/40 | Loss=0.5029 | AUC=0.9805 | Dice=0.8234 | Threshold=0.75 | Accuracy=0.9572 | Sensitivity=0.8309 | Specificity=0.9744 | Precision=0.8160 | 3.0 min
fold_4 | Epoch 31/40 | Loss=0.4974 | AUC=0.9815 | Dice=0.8253 | Threshold=0.75 | Accuracy=0.9578 | Sensitivity=0.8299 | Specificity=0.9753 | Precision=0.8206 | 3.1 min

fold_4 | Epoch 32/40 | Loss=0.5017 | AUC=0.9812 | Dice=0.8238 | Threshold=0.80 |
Accuracy=0.9575 | Sensitivity=0.8286 | Specificity=0.9750 | Precision=0.8190 | 3.2 min
fold_4 | Epoch 33/40 | Loss=0.4987 | AUC=0.9812 | Dice=0.8240 | Threshold=0.70 |
Accuracy=0.9569 | Sensitivity=0.8419 | Specificity=0.9725 | Precision=0.8069 | 3.3 min
fold_4 | Epoch 34/40 | Loss=0.4993 | AUC=0.9811 | Dice=0.8230 | Threshold=0.75 |
Accuracy=0.9574 | Sensitivity=0.8259 | Specificity=0.9753 | Precision=0.8202 | 3.4 min
fold_4 | Epoch 35/40 | Loss=0.4859 | AUC=0.9810 | Dice=0.8243 | Threshold=0.70 |
Accuracy=0.9571 | Sensitivity=0.8385 | Specificity=0.9733 | Precision=0.8107 | 3.5 min
fold_4 | Epoch 36/40 | Loss=0.4957 | AUC=0.9812 | Dice=0.8247 | Threshold=0.70 |
Accuracy=0.9573 | Sensitivity=0.8379 | Specificity=0.9735 | Precision=0.8119 | 3.6 min
fold_4 | Epoch 37/40 | Loss=0.4899 | AUC=0.9812 | Dice=0.8239 | Threshold=0.75 |
Accuracy=0.9575 | Sensitivity=0.8288 | Specificity=0.9751 | Precision=0.8192 | 3.7 min
fold_4 | Epoch 38/40 | Loss=0.4892 | AUC=0.9808 | Dice=0.8241 | Threshold=0.70 |
Accuracy=0.9571 | Sensitivity=0.8371 | Specificity=0.9735 | Precision=0.8114 | 3.8 min
fold_4 | Epoch 39/40 | Loss=0.4941 | AUC=0.9811 | Dice=0.8242 | Threshold=0.75 |
Accuracy=0.9576 | Sensitivity=0.8284 | Specificity=0.9752 | Precision=0.8201 | 3.9 min
Early stopping: validation Dice did not improve for 8 epochs.
Saved checkpoint: /content/vessel_model_5fold/fold_4_best.pt
Saved report: /content/vessel_model_5fold/fold_4_report.json

==========
RUN: fold_5
Training IDs: [21, 22, 23, 26, 27, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40]
Validation IDs: [24, 25, 28, 36]
Normalization: mean=0.4312, std=0.1070
Training vessel fraction: 0.1188
==========
fold_5 | Epoch 01/40 | Loss=1.6264 | AUC=0.9105 | Dice=0.6973 | Threshold=0.65 |
Accuracy=0.9069 | Sensitivity=0.7070 | Specificity=0.9426 | Precision=0.6877 | 0.1 min
fold_5 | Epoch 02/40 | Loss=1.4669 | AUC=0.9405 | Dice=0.7494 | Threshold=0.70 |
Accuracy=0.9272 | Sensitivity=0.7176 | Specificity=0.9647 | Precision=0.7841 | 0.2 min
fold_5 | Epoch 03/40 | Loss=1.3723 | AUC=0.9526 | Dice=0.7850 | Threshold=0.75 |
Accuracy=0.9363 | Sensitivity=0.7665 | Specificity=0.9667 | Precision=0.8044 | 0.3 min
fold_5 | Epoch 04/40 | Loss=1.2536 | AUC=0.9557 | Dice=0.7919 | Threshold=0.70 |
Accuracy=0.9374 | Sensitivity=0.7847 | Specificity=0.9648 | Precision=0.7993 | 0.4 min
fold_5 | Epoch 05/40 | Loss=1.0784 | AUC=0.9605 | Dice=0.8057 | Threshold=0.75 |
Accuracy=0.9414 | Sensitivity=0.8005 | Specificity=0.9667 | Precision=0.8110 | 0.5 min
fold_5 | Epoch 06/40 | Loss=0.8736 | AUC=0.9628 | Dice=0.8063 | Threshold=0.65 |
Accuracy=0.9423 | Sensitivity=0.7920 | Specificity=0.9692 | Precision=0.8212 | 0.6 min
fold_5 | Epoch 07/40 | Loss=0.7183 | AUC=0.9636 | Dice=0.8142 | Threshold=0.40 |
Accuracy=0.9440 | Sensitivity=0.8085 | Specificity=0.9683 | Precision=0.8200 | 0.7 min

fold_5 | Epoch 08/40 | Loss=0.6380 | AUC=0.9705 | Dice=0.8192 | Threshold=0.50 | Accuracy=0.9451 | Sensitivity=0.8208 | Specificity=0.9673 | Precision=0.8177 | 0.8 min
fold_5 | Epoch 09/40 | Loss=0.6074 | AUC=0.9716 | Dice=0.8207 | Threshold=0.55 | Accuracy=0.9460 | Sensitivity=0.8154 | Specificity=0.9693 | Precision=0.8261 | 0.9 min
fold_5 | Epoch 10/40 | Loss=0.5830 | AUC=0.9723 | Dice=0.8251 | Threshold=0.60 | Accuracy=0.9470 | Sensitivity=0.8237 | Specificity=0.9691 | Precision=0.8265 | 1.0 min
fold_5 | Epoch 11/40 | Loss=0.5815 | AUC=0.9739 | Dice=0.8271 | Threshold=0.60 | Accuracy=0.9480 | Sensitivity=0.8204 | Specificity=0.9708 | Precision=0.8340 | 1.1 min
fold_5 | Epoch 12/40 | Loss=0.5690 | AUC=0.9752 | Dice=0.8298 | Threshold=0.65 | Accuracy=0.9488 | Sensitivity=0.8235 | Specificity=0.9712 | Precision=0.8362 | 1.2 min
fold_5 | Epoch 13/40 | Loss=0.5635 | AUC=0.9746 | Dice=0.8283 | Threshold=0.55 | Accuracy=0.9478 | Sensitivity=0.8294 | Specificity=0.9690 | Precision=0.8271 | 1.3 min
fold_5 | Epoch 14/40 | Loss=0.5542 | AUC=0.9736 | Dice=0.8275 | Threshold=0.70 | Accuracy=0.9479 | Sensitivity=0.8238 | Specificity=0.9701 | Precision=0.8312 | 1.4 min
fold_5 | Epoch 15/40 | Loss=0.5519 | AUC=0.9760 | Dice=0.8296 | Threshold=0.45 | Accuracy=0.9479 | Sensitivity=0.8360 | Specificity=0.9679 | Precision=0.8234 | 1.5 min
fold_5 | Epoch 16/40 | Loss=0.5529 | AUC=0.9753 | Dice=0.8308 | Threshold=0.50 | Accuracy=0.9485 | Sensitivity=0.8334 | Specificity=0.9691 | Precision=0.8282 | 1.6 min
fold_5 | Epoch 17/40 | Loss=0.5387 | AUC=0.9765 | Dice=0.8321 | Threshold=0.50 | Accuracy=0.9487 | Sensitivity=0.8380 | Specificity=0.9685 | Precision=0.8263 | 1.7 min
fold_5 | Epoch 18/40 | Loss=0.5287 | AUC=0.9765 | Dice=0.8342 | Threshold=0.55 | Accuracy=0.9501 | Sensitivity=0.8272 | Specificity=0.9721 | Precision=0.8413 | 1.8 min
fold_5 | Epoch 19/40 | Loss=0.5327 | AUC=0.9770 | Dice=0.8337 | Threshold=0.55 | Accuracy=0.9495 | Sensitivity=0.8352 | Specificity=0.9699 | Precision=0.8322 | 1.9 min
fold_5 | Epoch 20/40 | Loss=0.5333 | AUC=0.9767 | Dice=0.8335 | Threshold=0.50 | Accuracy=0.9500 | Sensitivity=0.8256 | Specificity=0.9722 | Precision=0.8415 | 2.0 min
fold_5 | Epoch 21/40 | Loss=0.5294 | AUC=0.9775 | Dice=0.8338 | Threshold=0.55 | Accuracy=0.9495 | Sensitivity=0.8343 | Specificity=0.9701 | Precision=0.8332 | 2.1 min
fold_5 | Epoch 22/40 | Loss=0.5253 | AUC=0.9768 | Dice=0.8342 | Threshold=0.60 | Accuracy=0.9497 | Sensitivity=0.8340 | Specificity=0.9704 | Precision=0.8344 | 2.2 min
fold_5 | Epoch 23/40 | Loss=0.5173 | AUC=0.9783 | Dice=0.8364 | Threshold=0.55 | Accuracy=0.9504 | Sensitivity=0.8364 | Specificity=0.9707 | Precision=0.8363 | 2.3 min
fold_5 | Epoch 24/40 | Loss=0.5234 | AUC=0.9773 | Dice=0.8337 | Threshold=0.50 | Accuracy=0.9498 | Sensitivity=0.8299 | Specificity=0.9712 | Precision=0.8375 | 2.4 min
fold_5 | Epoch 25/40 | Loss=0.5199 | AUC=0.9772 | Dice=0.8356 | Threshold=0.60 | Accuracy=0.9497 | Sensitivity=0.8431 | Specificity=0.9688 | Precision=0.8283 | 2.5 min
fold_5 | Epoch 26/40 | Loss=0.5148 | AUC=0.9779 | Dice=0.8360 | Threshold=0.55 | Accuracy=0.9502 | Sensitivity=0.8370 | Specificity=0.9704 | Precision=0.8349 | 2.6 min
fold_5 | Epoch 27/40 | Loss=0.5141 | AUC=0.9776 | Dice=0.8372 | Threshold=0.60 | Accuracy=0.9508 | Sensitivity=0.8339 | Specificity=0.9717 | Precision=0.8406 | 2.7 min
fold_5 | Epoch 28/40 | Loss=0.5075 | AUC=0.9776 | Dice=0.8372 | Threshold=0.50 | Accuracy=0.9502 | Sensitivity=0.8437 | Specificity=0.9693 | Precision=0.8308 | 2.8 min
fold_5 | Epoch 29/40 | Loss=0.5095 | AUC=0.9778 | Dice=0.8364 | Threshold=0.55 | Accuracy=0.9502 | Sensitivity=0.8397 | Specificity=0.9699 | Precision=0.8332 | 2.9 min

fold_5 | Epoch 30/40 | Loss=0.5060 | AUC=0.9775 | Dice=0.8369 | Threshold=0.55 | Accuracy=0.9504 | Sensitivity=0.8398 | Specificity=0.9701 | Precision=0.8341 | 3.0 min
fold_5 | Epoch 31/40 | Loss=0.5048 | AUC=0.9779 | Dice=0.8364 | Threshold=0.55 | Accuracy=0.9504 | Sensitivity=0.8361 | Specificity=0.9708 | Precision=0.8366 | 3.1 min
fold_5 | Epoch 32/40 | Loss=0.5010 | AUC=0.9778 | Dice=0.8371 | Threshold=0.60 | Accuracy=0.9506 | Sensitivity=0.8364 | Specificity=0.9711 | Precision=0.8378 | 3.2 min
fold_5 | Epoch 33/40 | Loss=0.5032 | AUC=0.9783 | Dice=0.8377 | Threshold=0.60 | Accuracy=0.9510 | Sensitivity=0.8338 | Specificity=0.9719 | Precision=0.8416 | 3.3 min
fold_5 | Epoch 34/40 | Loss=0.5019 | AUC=0.9786 | Dice=0.8363 | Threshold=0.60 | Accuracy=0.9503 | Sensitivity=0.8375 | Specificity=0.9704 | Precision=0.8351 | 3.4 min
fold_5 | Epoch 35/40 | Loss=0.4972 | AUC=0.9780 | Dice=0.8370 | Threshold=0.55 | Accuracy=0.9506 | Sensitivity=0.8366 | Specificity=0.9709 | Precision=0.8374 | 3.5 min
fold_5 | Epoch 36/40 | Loss=0.4951 | AUC=0.9778 | Dice=0.8366 | Threshold=0.55 | Accuracy=0.9504 | Sensitivity=0.8381 | Specificity=0.9704 | Precision=0.8351 | 3.6 min
fold_5 | Epoch 37/40 | Loss=0.4933 | AUC=0.9778 | Dice=0.8366 | Threshold=0.55 | Accuracy=0.9499 | Sensitivity=0.8450 | Specificity=0.9687 | Precision=0.8284 | 3.7 min
fold_5 | Epoch 38/40 | Loss=0.4953 | AUC=0.9781 | Dice=0.8369 | Threshold=0.55 | Accuracy=0.9504 | Sensitivity=0.8395 | Specificity=0.9702 | Precision=0.8344 | 3.8 min
fold_5 | Epoch 39/40 | Loss=0.4937 | AUC=0.9782 | Dice=0.8366 | Threshold=0.55 | Accuracy=0.9504 | Sensitivity=0.8377 | Specificity=0.9705 | Precision=0.8356 | 3.9 min
fold_5 | Epoch 40/40 | Loss=0.4957 | AUC=0.9780 | Dice=0.8370 | Threshold=0.50 | Accuracy=0.9502 | Sensitivity=0.8431 | Specificity=0.9693 | Precision=0.8309 | 4.0 min
Saved checkpoint: /content/vessel_model_5fold/fold_5_best.pt
Saved report: /content/vessel_model_5fold/fold_5_report.json

==========
FIVE-FOLD CROSS-VALIDATION RESULTS
==========

Fold	Dice	AUC	Accuracy	Sensitivity	Specificity	Precision
fold_1	0.8111	0.9794	0.9554	0.8073	0.9753	0.8149
fold_2	0.8276	0.9840	0.9576	0.8451	0.9730	0.8109
fold_3	0.7923	0.9733	0.9508	0.8070	0.9697	0.7781
fold_4	0.8253	0.9815	0.9578	0.8299	0.9753	0.8206
fold_5	0.8377	0.9783	0.9510	0.8338	0.9719	0.8416

MEAN ∓ STANDARD DEVIATION
Dice : 0.8188 ∓ 0.0176
Auc : 0.9793 ∓ 0.0040
Accuracy : 0.9545 ∓ 0.0035
Sensitivity : 0.8246 ∓ 0.0169
Specificity : 0.9731 ∓ 0.0024
Precision : 0.8132 ∓ 0.0229

Cross-validation summary saved to: /content/vessel_model_5fold/five_fold_cv_summary.json

Training final model using all 20 labeled images for 31 epochs.

```
================================================================================
RUN: final_all20
Training IDs: [21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40]
Validation IDs: None — final model
Normalization: mean=0.4358, std=0.1060
Training vessel fraction: 0.1254
================================================================================
final_all20 | Epoch 01/31 | Loss=1.4273 | 0.1 min
final_all20 | Epoch 02/31 | Loss=1.2011 | 0.2 min
final_all20 | Epoch 03/31 | Loss=1.0000 | 0.3 min
final_all20 | Epoch 04/31 | Loss=0.7914 | 0.3 min
final_all20 | Epoch 05/31 | Loss=0.6681 | 0.4 min
final_all20 | Epoch 06/31 | Loss=0.6298 | 0.5 min
final_all20 | Epoch 07/31 | Loss=0.5981 | 0.6 min
final_all20 | Epoch 08/31 | Loss=0.5852 | 0.7 min
final_all20 | Epoch 09/31 | Loss=0.5821 | 0.8 min
final_all20 | Epoch 10/31 | Loss=0.5740 | 0.9 min
final_all20 | Epoch 11/31 | Loss=0.5653 | 1.0 min
final_all20 | Epoch 12/31 | Loss=0.5560 | 1.0 min
final_all20 | Epoch 13/31 | Loss=0.5448 | 1.1 min
final_all20 | Epoch 14/31 | Loss=0.5526 | 1.2 min
final_all20 | Epoch 15/31 | Loss=0.5445 | 1.3 min
final_all20 | Epoch 16/31 | Loss=0.5373 | 1.4 min
final_all20 | Epoch 17/31 | Loss=0.5412 | 1.5 min
final_all20 | Epoch 18/31 | Loss=0.5352 | 1.6 min
final_all20 | Epoch 19/31 | Loss=0.5308 | 1.7 min
final_all20 | Epoch 20/31 | Loss=0.5308 | 1.7 min
final_all20 | Epoch 21/31 | Loss=0.5242 | 1.8 min
final_all20 | Epoch 22/31 | Loss=0.5213 | 1.9 min
final_all20 | Epoch 23/31 | Loss=0.5220 | 2.0 min
final_all20 | Epoch 24/31 | Loss=0.5120 | 2.1 min
final_all20 | Epoch 25/31 | Loss=0.5107 | 2.2 min
final_all20 | Epoch 26/31 | Loss=0.5066 | 2.3 min
final_all20 | Epoch 27/31 | Loss=0.5040 | 2.3 min
final_all20 | Epoch 28/31 | Loss=0.5002 | 2.4 min
final_all20 | Epoch 29/31 | Loss=0.4993 | 2.5 min
final_all20 | Epoch 30/31 | Loss=0.5027 | 2.6 min
```

final_all20 | Epoch 31/31 | Loss=0.5038 | 2.7 min
Saved checkpoint: /content/vessel_model_5fold/final_all20_best.pt
Saved report: /content/vessel_model_5fold/final_all20_report.json

==
GLUCOIRIS TRAINING PIPELINE COMPLETED
==
Output directory: /content/vessel_model_5fold
CV summary: /content/vessel_model_5fold/five_fold_cv_summary.json
Fold checkpoints:
  /content/vessel_model_5fold/fold_1_best.pt
  /content/vessel_model_5fold/fold_2_best.pt
  /content/vessel_model_5fold/fold_3_best.pt
  /content/vessel_model_5fold/fold_4_best.pt
  /content/vessel_model_5fold/fold_5_best.pt
Final model: /content/vessel_model_5fold/final_all20_best.pt
Final report: /content/vessel_model_5fold/final_all20_report.json

IMPORTANT: The five-fold metrics estimate performance across the 20 labeled training images.
Evaluate on an independent test set before claiming final test performance.

Summary file structure:

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{
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  "method": "5-fold cross-validation",
  "images_per_fold": {
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    "validation": 4
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  "seed": 7,
  "folds": [
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      "train_ids": [
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        25,
        27,
        28,
        29,
        30,
        31,
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      34,
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      36,
      37,
      39,
      40
    ],
    "validation_ids": [
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      26,
      38
    ],
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      "threshold": 0.75,
      "dice": 0.8110593909483483,
      "accuracy": 0.9554359675877904,
      "sensitivity": 0.8072790197884395,
      "specificity": 0.9753495677139734,
      "precision": 0.8148753345541626,
      "tp": 86772,
      "fp": 19713,
      "fn": 20715,
      "tn": 779989
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    "checkpoint": "/content/vessel_model_5fold/fold_1_best.pt",
    "report": "/content/vessel_model_5fold/fold_1_report.json"
  },
  {
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    "train_ids": [
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      29,
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      31,
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      37,
      38,
      40
    ],
    "validation_ids": [
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      27,
      32,
      39
    ],
    "best_epoch": 34,
    "metrics": {
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      "threshold": 0.8,
      "dice": 0.8276277139494983,
      "accuracy": 0.9576322267975227,
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      "tn": 775318
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    "checkpoint": "/content/vessel_model_5fold/fold_2_best.pt",
    "report": "/content/vessel_model_5fold/fold_2_report.json"
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  {
    "run": "fold_3",
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```

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    "threshold": 0.8,
    "dice": 0.792331141744888,
    "accuracy": 0.9507549542670347,
    "sensitivity": 0.8070437445650119,
    "specificity": 0.9696874606975507,
    "precision": 0.7781453634085213,
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    "fn": 20414,
    "tn": 778724
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   "report": "/content/vessel_model_5fold/fold_3_report.json"
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  {
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    31,
    32,
    34,
```

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      "threshold": 0.75,
      "dice": 0.825265939272454,
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      "sensitivity": 0.8299354011087186,
      "specificity": 0.9752672869731882,
      "precision": 0.820648727009151,
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      "fp": 19795,
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      "tn": 780562
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    "report": "/content/vessel_model_5fold/fold_4_report.json"
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      30,
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      32,
      33,
      34,
      35,
```

```
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      39,
      40
    ],
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      25,
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      36
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    "best_epoch": 33,
    "metrics": {
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      "threshold": 0.6,
      "dice": 0.8377046076775209,
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      "precision": 0.8416020482809071,
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      "fp": 21653,
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      "tn": 750007
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    "report": "/content/vessel_model_5fold/fold_5_report.json"
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 "summary_mean_std": {
  "dice": {
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    "std": 0.0175900437491868,
    "fold_values": [
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      0.8276277139494983,
      0.792331141744888,
      0.825265939272454,
      0.8377046076775209
    ]
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    "mean": 0.9792956189975672,
    "std": 0.0039963552709974 5,
```

```
      "fold_values": [
        0.9794118935693571,
        0.9840059886683126,
        0.973315714960
```

```
============================================================
predict_full
Signature: (model, image, mean, std, device)
@torch.no_grad()
def predict_full(model, image, mean, std, device):
    model.eval()

    height, width = image.shape

    # Pad dimensions to multiples of 16.
    pad_h = (16 - height % 16) % 16
    pad_w = (16 - width % 16) % 16

    padded = np.pad(
        normalise(image, mean, std),
        ((0, pad_h), (0, pad_w)),
        mode="reflect"
    )

    tensor = torch.from_numpy(
        padded.copy()
    )[None, None].float().to(device)

    # Normal prediction.
    p1 = torch.sigmoid(model(tensor))[0, 0]
    p1 = p1[:height, :width]

    # Horizontal-flip test-time augmentation.
    flipped = torch.flip(tensor, dims=[-1])
    p2 = torch.sigmoid(model(flipped))[0, 0]
    p2 = torch.flip(p2, dims=[-1])
    p2 = p2[:height, :width]

    return ((p1 + p2) / 2).cpu().numpy()


============================================================
segmentation_metrics
Signature: (probabilities, labels, seed=0)
```

```
def segmentation_metrics(probabilities, labels, seed=0):
    labels = labels.astype(bool)

    # Use a reproducible subset of pixels for AUC.
    rng = np.random.default_rng(seed)

    indices = rng.choice(
        len(probabilities),
        size=min(len(probabilities), AUC_MAX_PIXELS),
        replace=False
    )

    auc = calculate_auc(
        probabilities[indices],
        labels[indices]
    )

    best = None

    # Select threshold by Dice on the validation data.
    for threshold in np.round(np.arange(0.20, 0.81, 0.05), 2):
        pred = probabilities >= threshold

        tp = int(np.count_nonzero(pred & labels))
        fp = int(np.count_nonzero(pred & ~labels))
        fn = int(np.count_nonzero(~pred & labels))
        tn = int(np.count_nonzero(~pred & ~labels))

        dice = 2 * tp / max(1, 2 * tp + fp + fn)

        precision = tp / max(1, tp + fp)
        sensitivity = tp / max(1, tp + fn)
        specificity = tn / max(1, tn + fp)
        accuracy = (tp + tn) / max(1, tp + tn + fp + fn)

        metrics = {
            "threshold": float(threshold),
            "dice": float(dice),
            "accuracy": float(accuracy),
            "sensitivity": float(sensitivity),
            "specificity": float(specificity),
            "precision": float(precision),
            "tp": tp,
            "fp": fp,
```

```
            "fn": fn,
            "tn": tn
        }

        if best is None or metrics["dice"] > best["dice"]:
            best = metrics

    return {"auc": float(auc), **best}


============================================================
FOLD 1 CHECKPOINT METADATA
{'run': 'fold_1',
 'model': 'U-Net',
 'parameters': {'base_channels': 16,
                'depth': 4,
                'dropout': 0.1,
                'patch_size': 96,
                'batch_size': 16,
                'learning_rate': 0.002,
                'weight_decay': 0.0001,
                'epochs_requested': 40,
                'iterations_per_epoch': 150},
 'train_ids': [21, 24, 25, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 39, 40],
 'validation_ids': [22, 23, 26, 38],
 'normalisation': {'mean': 0.43334098160266876, 'std': 0.10602119332179427},
 'best_epoch': 20,
 'validation_metrics': {'auc': 0.9794118935693571,
                        'threshold': 0.75,
                        'dice': 0.8110593909483483,
                        'accuracy': 0.9554359675877904,
                        'sensitivity': 0.8072790197884395,
                        'specificity': 0.9753495677139734,
                        'precision': 0.8148753345541626,
                        'tp': 86772,
                        'fp': 19713,
                        'fn': 20715,
                        'tn': 779989},
 'augmentation': ['horizontal flip',
                  'vertical flip',
                  '90-degree rotations',
                  'gamma',
                  'brightness',
                  'contrast',
```

'mild Gaussian noise']}

```
============================================================
DEVICE: cuda
load_drive_data: Not defined
load_dataset: Not defined
read_image: Not defined
preprocess_image: Not defined
load_image: Not defined
images: Not defined
vessels: Not defined
fovs: Not defined
predict_full: Available (function)
```